```
➤   def process(request)
      endpoint_description = Endpoint.description_of(request)
➤     @reporter.increment("api.requests.#{endpoint_description}")
    end
  end
```

Now, we just need to pass in the reporter collaborator when we *create* an APIRequestTracker instance, rather than passing it as an argument to process:

15-using-test-doubles-effectively/16/api_request_tracker/spec/api_request_tracker_spec.rb

```
it 'increments the request counter' do
  reporter = class_double(MetricsReporter)
  expect(reporter).to receive(:increment).with('api.requests.get_users')

➤ APIRequestTracker.new(reporter: reporter).process(request)
end
```

Constructor injection is our go-to technique for providing test doubles to our code. It's simple and explicit, and nicely documents what collaborators a class depends upon in the constructor. We should add that this style is not everyone's cup of tea. For a nuanced look at the advantages and disadvantages of dependency injection, see Tom Stuart's "How Testability Can Help" blog post.[5]

Sometimes, the constructor isn't available for us to modify. For example, web frameworks like Ruby on Rails often control object lifetimes, including the arguments to constructors. In these situations, we can fall back to *setter injection*:

15-using-test-doubles-effectively/17/api_request_tracker/lib/api_request_tracker.rb

```
class APIRequestTracker
➤ attr_writer :reporter
➤ def reporter
➤   @reporter ||= MetricsReporter.new
➤ end

  def process(request)
    endpoint_description = Endpoint.description_of(request)
➤   reporter.increment("api.requests.#{endpoint_description}")
  end
end
```

Here we've exposed a *setter* (via attr_writer) for our collaborator that can be used from our test to inject the dependency:

15-using-test-doubles-effectively/17/api_request_tracker/spec/api_request_tracker_spec.rb

```
it 'increments the request counter' do
  reporter = class_double(MetricsReporter)
  expect(reporter).to receive(:increment).with('api.requests.get_users')
```

5. http://codon.com/how-testability-can-help